

# Pinesphere Stay

## Functional Design Document

### Modules: Check-In, Check-Out & Payments

This document summarizes the workflows, fields, validations, database entities, user permissions, and future enhancements for the core operational modules of Pinesphere Stay.

#### 1. Overall Workflow

Booking Created  
↓  
Guest Arrival  
↓  
Check-In  
↓  
Stay Period  
↓  
Additional Charges (Restaurant/Laundry/etc.)  
↓  
Check-Out  
↓  
Payment Settlement  
↓  
Invoice Generated  
↓  
Room Available

#### 2. Check-In Module

##### Objective

Register guest, verify identity, assign room, collect advance, generate invoice, and mark room occupied.

##### Workflow

Guest Arrives → Search Booking/Walk-in → Guest Details → Verify Identity → Assign Room → Collect Advance → Generate Check-In → Invoice → Room Occupied → WhatsApp Welcome

##### Fields

- Booking ID, Booking Type, Status, Arrival Date, Departure Date, Nights
- Adults, Children, Total Guests
- Guest ID, Full Name, Mobile, Email, Address, City, State, Country

- Nationality, DOB, Gender
- ID Type, ID Number, Front/Back Image, OCR Result, Passport, Visa
- Room Number, Room Type, Floor, Occupancy Limit, Extra Bed
- Room Rent, Deposit, Discount, Taxes, Total Payable, Advance Paid, Pending Amount
- Guest Preferences, Notes, Vehicle Number, Emergency Contact

#### Actions

- Save
- Save Offline
- Print Registration
- Print Invoice
- WhatsApp Confirmation
- Cancel Check-In

#### Validations

- Mandatory mobile and ID
- Room availability
- Occupancy limit
- Duplicate guest detection
- Deposit validation
- Offline storage

#### Database Entities

- Guests
- Bookings
- CheckIns
- Rooms
- RoomAssignments
- IdentityProofs
- Invoices
- Payments
- GuestPreferences

### 3. Check-Out Module

#### Objective

Finalize stay, calculate all charges, collect payments, generate invoice, and release room.

#### Workflow

Search Guest → Active Stay → Calculate Bills → Restaurant/Laundry/Mini Bar/Damage → Taxes & Discounts → Payment → Invoice → Room Dirty → Housekeeping → Available

#### Billing Fields

- Room Charges

- Restaurant Bills
- Laundry
- Mini Bar
- Damage Charges
- Miscellaneous
- Discount
- GST
- Total

#### **Payment Summary**

- Advance Paid
- Remaining Balance
- Refund Amount
- Payment Status

#### **Database Entities**

- CheckOut
- Invoices
- InvoiceItems
- PaymentTransactions
- RoomStatus
- Housekeeping
- Feedback

### **4. Payments Module**

#### **Objective**

Manage every financial transaction during the guest lifecycle.

#### **Payment Sources**

- Booking Advance
- Deposit
- Room Rent
- Restaurant
- Laundry
- Mini Bar
- Extra Bed
- Damage Charges
- Other Services

#### **Payment Methods**

- Cash
- UPI
- Credit Card

- Debit Card
- Net Banking
- Wallet
- Split Payment

#### Payment Fields

- Transaction ID
- Invoice Number
- Guest Name
- Booking ID
- Payment Date
- Payment Mode
- Amount
- Reference Number
- UPI ID
- Bank Name
- Card Last 4 Digits
- Collected By
- Remarks

#### Payment Status

- Pending
- Partially Paid
- Fully Paid
- Refunded
- Cancelled

#### Database Entities

- Payments
- PaymentMethods
- PaymentTransactions
- Invoices
- Receipts
- Refunds
- CashRegister

### 5. Offline Flow

Collect Payment → Store Locally → Generate Receipt → Internet Available → Background Sync → Cloud Updated

### 6. Integrations

- Check-In: Booking, Guest, Room Management, OCR, WhatsApp, Invoice, Payments
- Check-Out: Housekeeping, Maintenance, Restaurant, Laundry, Inventory, Feedback

- Payments: Accounting, GST, Reports, Refunds, Analytics

## 7. User Permissions

| Action             | Owner | Manager | Reception | Accountant |
|--------------------|-------|---------|-----------|------------|
| Create Check-In    | ✓     | ✓       | ✓         | X          |
| Collect Payments   | ✓     | ✓       | ✓         | ✓          |
| Refund Payment     | ✓     | Limited | X         | ✓          |
| Generate Invoice   | ✓     | ✓       | ✓         | ✓          |
| Complete Check-Out | ✓     | ✓       | ✓         | X          |

## 8. Future Enhancements

- AI room assignment
- Face recognition
- OCR ID extraction
- Dynamic pricing
- Self Check-In via Guest Portal
- QR room access
- Automated refunds
- Revenue analytics

# Pinesphere Stay - Database Planning (MVP)

## Guests

| Column              | Format    | Notes                      |
|---------------------|-----------|----------------------------|
| guest_id            | UUID (PK) | Unique guest identifier    |
| full_name           | TEXT      | Guest full name            |
| mobile              | TEXT      | Primary contact            |
| email               | TEXT      | Optional                   |
| address             | TEXT      | Address                    |
| id_type             | ENUM      | Aadhaar/Passport/DL        |
| id_number           | TEXT      | ID number                  |
| verification_status | ENUM      | Pending/Reception Verified |
| created_at          | DATETIME  | Created timestamp          |

## Bookings

| Column        | Format    | Notes          |
|---------------|-----------|----------------|
| booking_id    | UUID (PK) | Unique booking |
| guest_id      | UUID (FK) | Guest          |
| room_id       | UUID (FK) | Room           |
| checkin_date  | DATE      | Arrival        |
| checkout_date | DATE      | Departure      |
| status        | ENUM      | Booking status |

## Rooms

| Column          | Format    | Notes          |
|-----------------|-----------|----------------|
| room_id         | UUID (PK) | Unique room    |
| room_number     | TEXT      | Display number |
| room_type       | TEXT      | Category       |
| capacity        | INTEGER   | Occupancy      |
| price_per_night | DECIMAL   | Tariff         |
| status          | ENUM      | Availability   |

## CheckIns

| Column     | Format    | Notes     |
|------------|-----------|-----------|
| checkin_id | UUID (PK) | Unique    |
| booking_id | UUID (FK) | Booking   |
| staff_id   | UUID (FK) | Reception |

|               |          |         |
|---------------|----------|---------|
| deposit       | DECIMAL  | Deposit |
| checked_in_at | DATETIME | Time    |

## CheckOuts

| Column        | Format    | Notes      |
|---------------|-----------|------------|
| checkout_id   | UUID (PK) | Unique     |
| checkin_id    | UUID (FK) | Check-in   |
| checkout_time | DATETIME  | Departure  |
| damage_charge | DECIMAL   | Damage fee |

## Invoices

| Column         | Format    | Notes        |
|----------------|-----------|--------------|
| invoice_id     | UUID (PK) | Unique       |
| booking_id     | UUID (FK) | Booking      |
| invoice_number | TEXT      | Invoice no   |
| grand_total    | DECIMAL   | Final amount |

## Payments

| Column           | Format    | Notes          |
|------------------|-----------|----------------|
| payment_id       | UUID (PK) | Unique         |
| invoice_id       | UUID (FK) | Invoice        |
| transaction_id   | TEXT      | Internal ID    |
| reference_number | TEXT      | UTR/POS Ref    |
| payment_method   | ENUM      | Cash/UPI/Card  |
| amount           | DECIMAL   | Received       |
| status           | ENUM      | Payment status |

## IdentityProofs

| Column      | Format    | Notes           |
|-------------|-----------|-----------------|
| proof_id    | UUID (PK) | Unique          |
| guest_id    | UUID (FK) | Guest           |
| image_path  | TEXT      | Encrypted image |
| ocr_json    | JSON      | OCR output      |
| verified_by | UUID (FK) | Staff           |

## Staff

| Column | Format | Notes |
|--------|--------|-------|
|--------|--------|-------|

|          |           |          |
|----------|-----------|----------|
| staff_id | UUID (PK) | Employee |
| name     | TEXT      | Name     |
| role     | ENUM      | Role     |
| mobile   | TEXT      | Phone    |

## AuditLogs

| Column    | Format    | Notes  |
|-----------|-----------|--------|
| log_id    | UUID (PK) | Unique |
| user_id   | UUID      | Actor  |
| action    | TEXT      | Action |
| entity    | TEXT      | Module |
| timestamp | DATETIME  | Time   |



# Pinesphere Stay

## Housekeeping & Maintenance Module Plan + Database Plan

**Module Objective:** Manage room cleaning, inspections, maintenance requests, task assignments, and room readiness while working completely offline.

### Workflow

Guest Checks Out → Room marked Dirty → Housekeeping Task Created → Staff Assigned → Cleaning Checklist Completed → Inspection → If issue found → Maintenance Ticket → Technician Assigned → Repair Completed → Final Inspection → Room Available

### Housekeeping Features

- Room cleaning queue
- Assign housekeeping staff
- Cleaning checklist
- Photo before/after cleaning
- Lost & Found
- Laundry status
- Cleaning remarks
- Room ready notification
- Offline task management

### Maintenance Features

- Create maintenance ticket
- Priority (Low/Medium/High/Critical)
- Electrical, Plumbing, AC, TV, Furniture categories
- Assign technician
- Repair status tracking
- Cost tracking
- Photo attachments
- Room blocking until repaired

### Table: HousekeepingTasks

| Column            | Format    | Notes                                   |
|-------------------|-----------|-----------------------------------------|
| task_id           | UUID (PK) | Unique task                             |
| room_id           | UUID (FK) | Room reference                          |
| assigned_staff_id | UUID (FK) | Housekeeping staff                      |
| status            | ENUM      | Pending/In Progress/Completed/Inspected |

|                  |          |                            |
|------------------|----------|----------------------------|
| priority         | ENUM     | Low/Medium/High            |
| checklist_status | JSON     | Cleaning checklist results |
| remarks          | TEXT     | Cleaner notes              |
| before_photo     | TEXT     | Encrypted image path       |
| after_photo      | TEXT     | Encrypted image path       |
| completed_at     | DATETIME | Completion time            |

**Table: MaintenanceTickets**

| Column            | Format    | Notes                            |
|-------------------|-----------|----------------------------------|
| ticket_id         | UUID (PK) | Unique ticket                    |
| room_id           | UUID (FK) | Affected room                    |
| reported_by       | UUID (FK) | Staff who reported               |
| assigned_to       | UUID (FK) | Technician                       |
| category          | ENUM      | Electrical/AC/Plumbing/etc.      |
| priority          | ENUM      | Low/Medium/High/Critical         |
| issue_description | TEXT      | Problem details                  |
| status            | ENUM      | Open/In Progress/Resolved/Closed |
| repair_cost       | DECIMAL   | Repair expense                   |
| created_at        | DATETIME  | Created time                     |
| resolved_at       | DATETIME  | Completion time                  |

**Table: LostAndFound**

| Column      | Format    | Notes                    |
|-------------|-----------|--------------------------|
| item_id     | UUID (PK) | Unique item              |
| room_id     | UUID (FK) | Found in room            |
| description | TEXT      | Item details             |
| found_by    | UUID (FK) | Staff                    |
| status      | ENUM      | Stored/Returned/Disposed |
| photo       | TEXT      | Image path               |

**Table: RoomInspection**

| Column        | Format    | Notes            |
|---------------|-----------|------------------|
| inspection_id | UUID (PK) | Inspection       |
| room_id       | UUID (FK) | Room             |
| inspected_by  | UUID (FK) | Supervisor       |
| result        | ENUM      | Pass/Fail        |
| remarks       | TEXT      | Inspection notes |

|              |          |           |
|--------------|----------|-----------|
| inspected_at | DATETIME | Timestamp |
|--------------|----------|-----------|

## Relationships

Rooms → HousekeepingTasks (1:M)  
Rooms → MaintenanceTickets (1:M)  
Rooms → RoomInspection (1:M)  
Staff → HousekeepingTasks (1:M)  
Staff → MaintenanceTickets (1:M)

## Status Flow

Vacant → Dirty → Cleaning → Inspection → Available  
Issue Found → Maintenance → Repaired → Available

## Offline Strategy

All tasks, photos, checklists, and maintenance tickets are stored in the encrypted local database and synchronized automatically when internet connectivity returns.